	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	582 - BuildWorldKernel	(1, 1, 1)x(1, 1, 1)	39.93 ms	86,850,452	0 - NVIDIA GeForce RTX 4060 Laptop GPU	2.17 Ghz	[5464] PathTracingCUDA.exe	

Summary Details Source Context Comments Raw Session

Compare Tools View Export

This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [%]	Function Name	Demangled Name	Duration [ms] (824.97 ms)	Runtime Improvement [ms] (38.87 ms)	Compute Throughput [%]	Memory Throughput [%]	# Registers [
0	96.88	**BuildWorldKernel**	**BuildWorldKernel..**	39.93	38.68	0.06	0.08	
1	96.88	BuildCameraKernel	BuildCameraKernel..	0.09	0.08	0.05	0.35	
2	0.00	RenderKernel	RenderKernel..Hitta...	784.85	0.00	29.21	58.95	
3	96.88	DestroyKernel	DestroyKernel..Hitt...	0.11	0.11	0.04	0.11	

The following performance optimization opportunities were discovered for this result. Follow the rule links to see more context on the Details page.
Note: *Speedup estimates provide upper bounds for the optimization potential of a kernel assuming its overall algorithmic structure is kept unchanged.*

Block Size
Est. Speedup: 96.88%

Threads are executed in groups of 32 threads called warps. This kernel launch is configured to execute 1 threads per block. Consequently, some threads in a warp are masked off and those hardware resources are unused. Try changing the number of threads per block to be a multiple of 32 threads. Between 128 and 256 threads per block is a good initial range for experimentation. Use smaller thread blocks rather than one large thread block per multiprocessor if latency affects performance. This is particularly beneficial to kernels that frequently call __syncthreads(). See the Hardware Model description for more details on launch configurations.

▸ Key Performance Indicators

Small Grid
Est. Speedup: 95.83%

The grid for this launch is configured to execute only 1 block, which is less than the GPU's 24 multiprocessors. This can underutilize some multiprocessors. If you do not intend to execute this kernel concurrently with other workloads, consider reducing the block size to have at least one block per multiprocessor or increase the size of the grid to fully utilize the available hardware resources. See the Hardware Model description for more details on launch configurations.